Beat Machine S3 – User Manual (Labelled PCB)

Modes / Inner Button	Bank	Colour	Click Action	Hold Click	Lock Click	Combo Hold Click/Turn	Dial A	Dial B	Dial C	Dial D	Dial E	Dial F	Dial G	Dial H
Global (no button pressed)	0		1-16: toggle step state.				**Filter (LP on Wooden, Dual on PCB)**	**Density**	**Evolve**	**Chaos**	**Drive**	**Tempo**	**Length**	**Volume PCB, HPF Wooden**
Mix	1	Red		1-8: Play sound			Volume for part 1	Vol 2	Vol 3	Vol 4	Vol 5	Vol 6	Vol 7	Vol 8
FX	2	Orange		1-16: Delay time			Reverb Level	Reverb Time	Delay Level	Delay Time	Accel X Amnt	Accel Y Amnt		LED brightness
Sequence	3	Light Green	Tap Tempo	1-16: Select factory pattern preset	1-16: Manual pattern edit	FX - 1-16: Select user pattern preset Mix - 1-16: Save user pattern preset Sound – 1-12: Diatonic transpose	Half or Double Time pattern	Step Size	Slice Amnt	Swing	Accent	Tempo Fine Control	Key (1-12, 1 is C)	Scale (11-16 – see p.7)
Generate	X	Blue	New Euclidean Kit Pattern			Mix – Two-three kit pattern FX – Probability Pattern Seq – Polyrhythm kit pattern Stutter – Init empty pattern Part – Euclidean part Part & Mix – Two-three part Part & FX – Probability Part Part & Seq – Polyrhythm part Sound – Generate Kit sound Part & Sound – Gen Part Sound								
Play	X	Green	Start/Stop											
Stutter	4	Purple		1-16: Select stutter step(s)		Generate – Init empty pattern Part – Stutter current part only	Temp Filter (LP on Wooden, Dual on PCB)	Temp Part Pitch change	Stutter Length	Temp Chaos		Global LPF	Curr Part tone decay	Curr Part noise decay
Part	5	Aqua		1-8: Select part 9-16: Mute Part Double click 9-16: Solo Part		Generate – Euclidean part Gen & Mix – Two-three part Gen & FX – Probability Part Gen & Seq – Polyrhythm part Gen & Sound – Gen Part Sound Stutter – Stutter current part	Part Half or Double Time pattern	Part Density	Part Evolve	Rotate	Part Octave	Part Tempo (& Seq fine control)	Part Length	Pitched/Unpitched + range
Sound	6	Pink		1-16: Select factory sound preset		MIX - 1-16: Select user sound preset FX - 1-16: Save user sound preset Seq – Dial2 - Pitch fine tune	Osc Balance	Pitch	Mod Rate	Mod Depth	Noise Cutoff	Attack	Tone Decay	Noise Decay

Quick Start Guide

Power on. Connect to a powered USB cable and connect audio to headphones or mixer.

UI Orientation

There are two rings of **LEDs** and **button** pairs. 16 in an outer ring and 8 in an inner ring. The outer ring shows and edits the rhythmic pattern. The inner ring relates to the BMs operating mode, or state, and these LEDs pair with 8 inner buttons.

The inner ring of **buttons** are labelled and trigger actions (e.g. play and stop or generate) or they change 'modes'. *Hold* labelled buttons to temporarily change mode or *double-click* to lock to a mode. The dials (A-H) change function in each mode.

Dials are marked A to H in clockwise order, starting at 12 o'clock. Dial adjustment triggers a yellow LED indicator of dial value on the outer LED ring. When modes change, dials change function. Dials need to pass the previous value (the latch target) before the parameter changes. When dials are moved the nearest outer LED is a latching indicator, if lit red the dial is higher that the target value, if lit blue the dial is lower that the target, when lit yellow the dial is active and parameter values are being updated.

Instant Gratification

Press PLAY (green LED) to start and stop a sequence.

Dial F adjusts tempo, … (see matrix chart), dial H adjusts Volume on the BM PCB. On BM Wooden, the centre dial adjusts the volume.

It's a **generative** drum machine, so generate a new pattern by pressing GENERATE (dark blue LED).

It's **probabilistic**, and in GLOBAL mode dial B controls step probability, from 0% on the left (drop out steps) through 100% in the middle (play pattern normally), and 200% on the right (fill in step gaps).

Manual Pattern Edit

In the default Global mode, the outer LEDs show a colour mixture of all parts. Double-click the SEQUENCE button (yellow LED) to **show** the current part pattern. Then press the outer ring of buttons (1-16) to toggle steps on and off. A single click on outer buttons **toggle steps** on/off in the current part. Double-clicking outer ring buttons will add two 32nd notes to that 16th note step. Clicking to turn off a step, will clear all notes in that 16th note block. Click the Sequence button again to return to Global mode.

Change **parts** by holding the PART button (G) to enter Part mode and press an outer button 1 – 8. LEDs will redraw to show the current part's pattern. Active steps in each part use a different colour, and non-active steps are a dull aqua colour. If patterns are less than 16 steps, LEDs on non-used steps are off. To stay in Part mode double-click the PART button. Click PART again to exit to Global mode.

Each of the 8 parts uses a different highlight **colour** for active steps on the outer ring of LEDs. The same colours are used on the inner ring of mode button LEDs. Starting with red on inner button and part 1, yellow next, and so on clockwise. Looking at the inner LEDs can be a reminder of which colour is used for which part. Be aware that the LEDs can show different shades of a 'colour' at different intensities.

Change part **volume** – hold the MIX button (A) to go into Mix mode and use dials A – H to adjust the volume of parts (voices/tracks/instruments) 1 – 8.

Mute parts by holding the PART button *and* press outer button 9 – 16. LEDs 9-16 will flash when their parts (1-8) are muted. Press again to unmute. Hold PART and double-click 9-16 to solo a part. Double-click again to exit solo.

Initialise an **empty pattern** by holding the EFFECTS and STUTTER buttons and clicking GENERATE. This also resets the pattern length to 16 steps and select part 1 as the active part. Now add your own steps to each part.

Generative Pattern Algorithms

Euclidean rhythms - generate a new Euclidean pattern by clicking the GENERATE button.

Polyrhythms – generate a new polyrhythm by holding the SEQUENCE button and clicking GENERATE.

2 and 3 step rhythms - generate a new Small Step Duration Pattern by holding the MIX button and clicking GENERATE.

Probabilistic rhythms – generate a new structured random pattern by holding the EFFECTS button and clicking GENERATE.

Generate Algorithmic Patterns for Each Part

For a new Euclidean pattern on the current part only, hold the PART button and click GENERATE. For a new 2-3 step part, hold the MIX button and click GENERATE. For a new Polyrhythm on the current part, hold the SEQUENCE button and click GENERATE.

Updating individual parts with generated rhythms, allows patterns to have a mix of rhythms generated by different algorithms and then edited manually to your liking.

Modes and Dials

The BM defaults to Global mode where the **dials** correspond to Filter, Density, Evolve, Chaos, Drive, Tempo, Pattern Length, and Volume, which effect all parts.

The dials can have other functions when in a different mode (see the matrix). Inner buttons other than PLAY and GENERATE are used to change mode. To temporarily enter a different mode, hold down an inner ring button then turn a dial to adjust the associated parameter. To lock the BM in a mode, double click an inner button, the associated LED will glow brighter. To go back to Global mode, click any inner button.

Change The Sounds

It’s a **multitimbral** synthesizer. All sounds are synthesized in real time. Each part uses the same synthesis engine but has its own sound settings.

Select a factory **sound preset** by holding the SOUND button and pressing outer buttons 1-16 (only some slots may have a factory preset).

Edit sounds by holding the SOUND button to enter Sound mode. Then use dials A-H to change sound parameters. Dial A changes the balance between tone and noise, dial B changes the tone pitch, .etc., see the matrix chart above. Double-clicking on the SOUND button will lock the BM in Sound mode for extended sound design sessions. Click the SOUND button again to exit back to Global mode.

All parts use the same synthesis engine, but by convention (and a bit by design) part 1 is a kick drum, part 2 is a snare, part 3 is hi hat, part 4 is a clap, part 5 is a clave, part 6 is a tom, part 7 is wooden percussion, and part 8 is metallic percussion. A synthesis architecture diagram is shown on the Synth Controls page of this user guide.

Generate sounds by holding the Sound button and clicking GENERATE. Randomised sounds are produced. These are oriented toward the part conventions (kick, snare, hats, etc). Select factory sound preset 1 to return to the startup sound set.

Sounds can be **pitched**. By default sounds are treated as untuned – in that they always play at the same pitch. Holding PART and moving dial H away from zero will make the sound on the current part pitched. Use of the GENERATE button will also produce a pitch contour for that part. The range of that contour for the current part is controlled by holding PART and moving dial H. Select a scale and key in SEQUENCE mode with dials G and H.

Go Nuts

Parts can be of different **lengths**, allowing for **polymetric** patterns. Hold the PART button and turn dial G to adjust the length of the current part. Go to a different part and repeat the process.

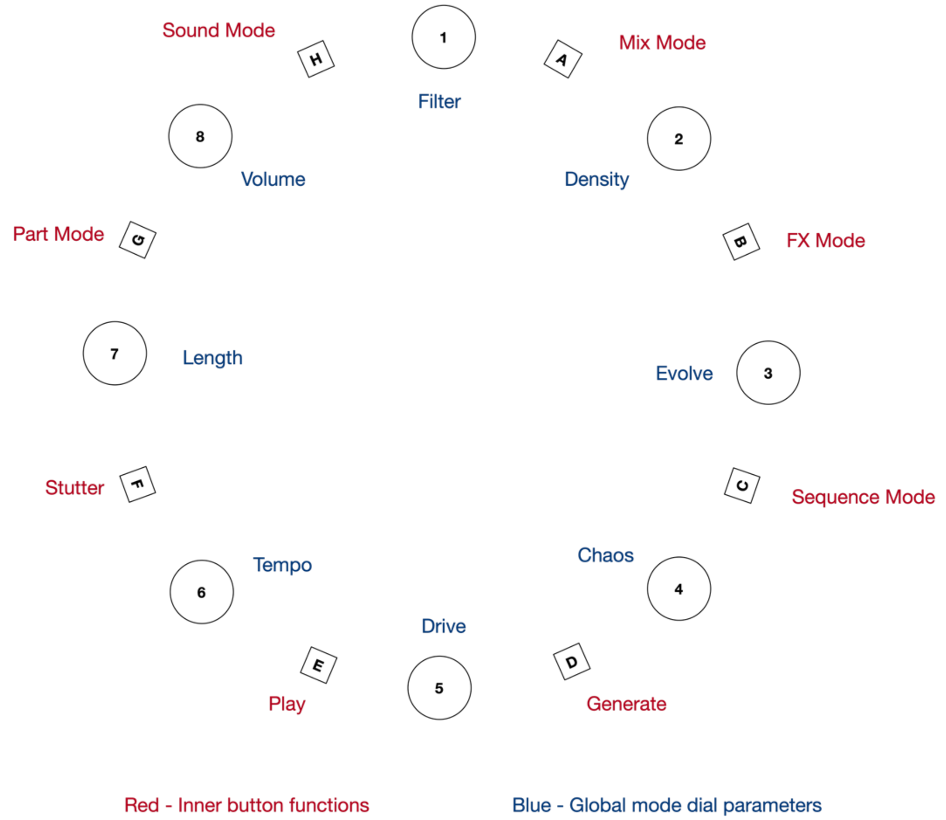
Rotate a part. Hold the PART button and turn dial D to shift the metrical phase of the current part. (Think of Reich’s ‘Clapping Music’).

Parts can be at different tempi, allowing for time **phasing** of patterns (Think of Reich’s ‘Piano Phase’). Hold the PART button and turn dial F to change the tempo of the current part in 10 BPM increments. For finer tempo adjustments (1 BPM increments), hold the SEQUENCE button and turn dial F. Change to another part and repeat for even more polymetric shenanigans.

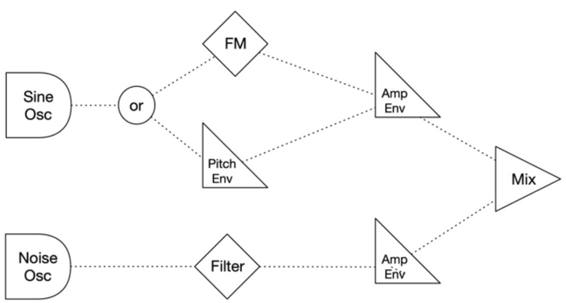
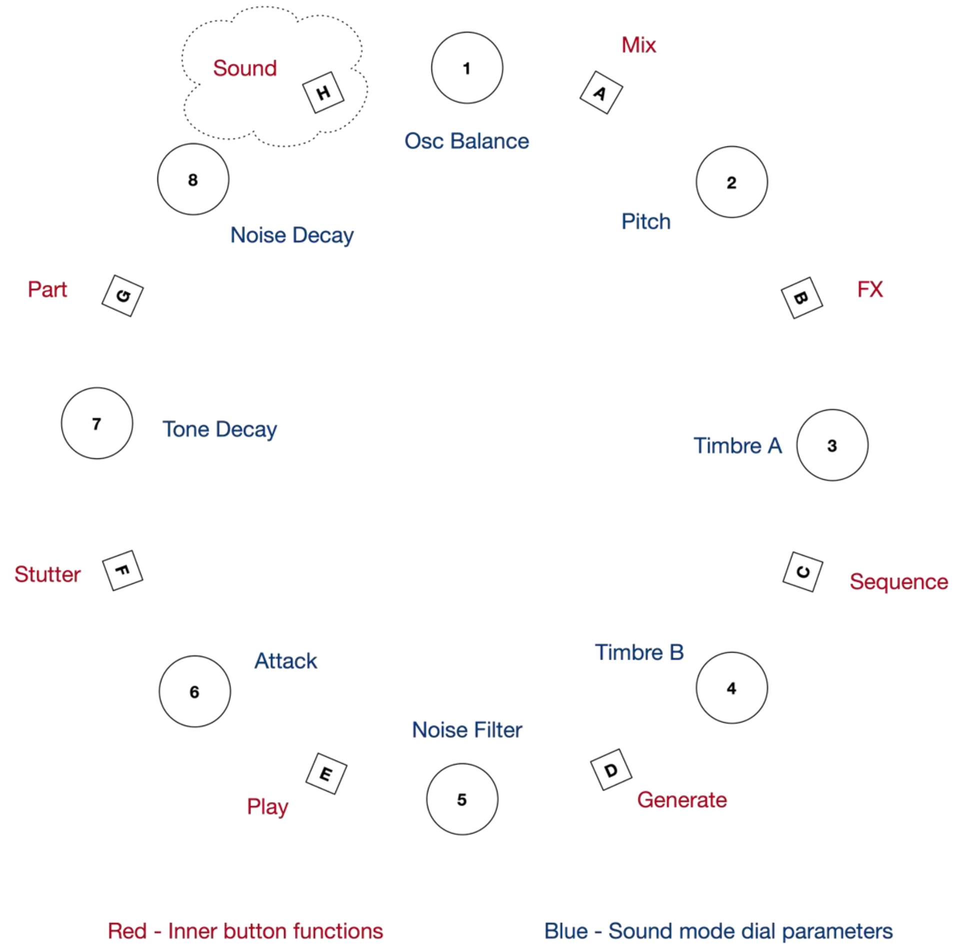
Update Firmware

The latest Beat Machine firmware can be installed via a web browser. Installation instructions and the current user manual can be accessed from the BM GitHub repository: https://github.com/algomusic/beatmachine

Global Mode Controls

Buttons	**Dials**	
• Play – Stop • Stutter • Part Select (PART+1:8) • Part Mute (PART+9:16) • Part Generate (PART+D) • Pattern Generate (GEN) • Polymetric Gen (SEQ+D) • 2&3 Step Gen (FX+D) • Probabilistic Gen (MIX+D) • Init Pattern (FX+STUT+GEN) • Sound Generate (SND+GEN)	1. Filter 2. Density 3. Evolve 4. Chaos 5. Drive 6. Tempo 7. Length 8. Volume	

Synthesis Controls

Buttons	Dials	
• Sound Mode	1. Oscillator Balance 2. Pitch 3. FM Ratio / Pitch Env On 4. FM Mod Depth / Pitch Env Depth 5. Noise Filter 6. Attack (tone & noise) 7. Tone Decay 8. Noise Decay Beat Machine S3 - Synthesis Architecture 	

Parameter details

Tempo – 120 BPM by default. When adjusting with dial F, BPM changes in 10 bpm increments at each of 16 steps, from 40 BPM at step 1 to 120 BPM at step 9, to 190 BPM at step 16. To fine tune the tempo in 1 BPM increments, hold the SEQUENCE button then turn dial F – this increases the base tempo up to 15 BPM. Change the tempo of the current part only by holding the PART button and moving dial F for 10 BPM increments or hold SEQUENCE & PART then move dial F to increase it in 1 BPM increments. Finally, **tap tempo** using the SEQUENCE button – 4 or more taps are required to set a tempo.

Density – Determines the probability that a step will sound, or not. Density defaults to 100% and ranges from 0% to 200% with 100% being at step 9.

Timbre – The purpose most sound synthesis controls are obvious from their names, however the two timbre controls have dual functionality. Timbre control A (Sound mode, dial C) switches between FM Ratio and Pitch Env On/Off. When dial C is at zero there is no FM and a **pitch envelope** on the tone is enabled. Timbre control B (Sound mode, dial D) then controls the **depth** of the pitch envelope; with 0 in the centre, to the left is increasing upward pitch bend, to the right is increasing downward pitch bend (typical of a TR-808 kick or Simmonds tom). When in Sound mode and dial C is above zero, this turns on FM and adjusts the Carrier:Modulator **ratio**. Dial D then adjusts the frequency modulation **depth**.

Scales

| 1: Maj Pentatonic | 2: Min Pentatonic | 3: Ionian | 4: Dorian |

| 5: Phrygian | 6: Lydian | 7: Mixolydian | 8: Aeolian |

| 9: Locrian | 10: Melodic Minor | 11: Harmonic Minor | 12: Whole Tone |

13: Blues	14: Japanese	15: Gypsy	16: Chromatic

MIDI – To turn MIDI on, restart the Beat Machine whilst holding down the 1 button on the outer ring. The Beat Machine will send note and sync data via USB-MIDI when patterns play. It will receive note data on channel 1 to trigger each voice. Mappings are:
Kick: 36 or 48, Snare: 38 or 49, HiHat: 42 or 50, Clap: 39 or 51, Clave: 41 or 52, Tom: 42 or 53, Wood: 45 or 54, and Metal: 47 or 55.

Factory Presets

There are 16 built-in preset patterns and 16 preset sounds. All presets include 8 parts each. Patterns and Sounds are independent, so you can mix and match as you like. Factory presets can't be overwritten so they are always available.

Factory Pattern Presets

1: 8 Beat	2: 8 Beat Fill	3: Rock	4: Rock Fill
5: Shuffle	6: Shuffle Fill	7: Funk	8: Funk Fill

9: Rap	10: Rap Fill	11: Triplet Rock	12: Jazz Swing
13: Bossa Nova	14: Reggae	15: Waltz	16: Clapping

Factory Sound Presets

1: BM Default	2: 909	3: 808	4: 707
5: 606	6:	7:	8:
9:	10:	11:	12:
13:	14:	15:	16:

User Presets

There are slots for 16 user preset patterns and 16 user preset sounds. All presets include 8 parts each. These user presets can be overwritten so you can update them as often as you like – for example, to put a collection together for a performance. Sound presets include the part volumes. Pattern presets include the sequence length(s), tempo(s) and density settings at the time of saving.

Preset Type	**Button Combo**	**Normal**	**With Part Held**
Load **User Sound**	Mix + Sound + Step	All 8 parts	Current part only
Load **User Pattern**	Mix + Sequence + Step	All 8 parts	Current part only
Load **Factory Sound**	Sound + Step	All 8 parts	Current part only

Load **Factory Pattern**	Sequence + Step	All 8 parts	Current part only
Save **User Sound**	FX + Sound + Step	All 8 parts	Current part only
Save **User Pattern**	FX + Sequence + Step	All 8 parts	Current part only
Delete **User Sound**	FX + Sound + Dbl-Step	All 8 parts	
Delete **User Pattern**	FX + Seq + Dbl-Step	All 8 parts	